

ZHOU WEIGHTS AND VALUATIVE SEPARATION BY GENERALIZED LELONG NUMBERS

SHIJIE BAO

ABSTRACT. We prove that Zhou weights are flat: their normalized generalized Lelong numbers equal their relative types on all plurisubharmonic germs. Their residual Monge–Ampère masses equal the volumes of the associated valuations. The proof combines a hypersurface multiplicity formula with tame approximation. A bounded-discrepancy refinement of the Bao–Guan–Zhou density construction then shows that generalized Lelong numbers tested against Zhou weights determine valutive transforms. This proves the remaining implication (4) $\Rightarrow$ (1) in Theorem A of Boucksom–Favre–Jonsson.

1. INTRODUCTION

Work at the origin $o \in \mathbb{C}^n$, with $n \geq 2$, and put $R = \mathcal{O}_{\mathbb{C}^n, o}$ and $\mathfrak{m} = \mathfrak{m}_o$. For a maximal plurisubharmonic weight Φ with isolated pole at o , write

$$\begin{aligned}\sigma(u, \Phi) &= \sup\{c \geq 0 : u \leq c\Phi + O(1)\}, \\ \nu_\Phi(u) &= (dd^c u \wedge (dd^c \Phi)^{n-1})\{o\}, \quad \tau_\Phi = (dd^c \Phi)^n\{o\}.\end{aligned}$$

Demailly’s comparison theorem gives $\tau_\Phi \sigma(u, \Phi) \leq \nu_\Phi(u)$. Equality for every psh germ is the defining property of a *flat* weight [13, Section 6]. Our first result establishes flatness for Zhou weights.

Zhou weights were introduced by Bao–Guan–Mi–Yuan [1] through maximality under a non-integrability constraint. They admit tame maximal representatives, and their holomorphic relative types define centered real valuations

$$w_\Phi(f) = \sigma(\log |f|, \Phi), \quad f \in R \setminus \{0\}.$$

Theorem 1.1 (Zhou weights are flat). *Let Φ be a local Zhou weight, represented by its tame maximal representative. Set $w = w_\Phi$, $\lambda = w(\mathfrak{m})$, and $v = w/\lambda$. Then*

$$(1.1) \quad \tau_\Phi = \text{vol}(w) > 0, \quad \nu_\Phi(u) = \text{vol}(w) \sigma(u, \Phi) \quad \text{for every psh germ } u.$$

In the normalized valutive space of Boucksom–Favre–Jonsson,

$$(1.2) \quad \widehat{\Phi}(v) = -\lambda^{-1}, \quad \text{MA}(\widehat{\Phi}) = \lambda \text{vol}(w) \delta_v, \quad \sigma(u, \Phi) = \lambda(-\widehat{u}(v)).$$

Date: September 7, 2026.

2020 Mathematics Subject Classification. 32U25, 13A18, 13H15, 14B05.

Key words and phrases. Zhou weight, Zhou valuation, generalized Lelong number, relative type, valuation volume, multiplier ideal, valutive transform.

For a centered real valuation w , we use the notation

$$\mathfrak{a}_t(w) = \{f \in R : w(f) \geq t\}, \quad \text{vol}(w) = \lim_{m \rightarrow \infty} \frac{n! \ell(R/\mathfrak{a}_m(w))}{m^n}.$$

The main algebraic input is

$$(1.3) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} = w(f) \text{vol}(w), \quad 0 \neq f \in \mathfrak{m}.$$

Here $e(I; S)$ denotes Hilbert–Samuel multiplicity. Cutkosky’s theorem [5] supplies the hypersurface limit; the colon identity $(\mathfrak{a}_t(w) : f) = \mathfrak{a}_{t-w(f)}(w)$ determines it by a telescoping sum. Tame approximation transfers the result to analytic masses. The valuative formulas of [4] then extend equality from holomorphic logarithms to all psh germs.

The second result concerns separation. The notation $\widehat{u}$ denotes the valuative transform of a psh germ in [4]; the sign convention is $\widehat{\log |f|}(v) = -v(f)$.

Theorem 1.2 (Separation by Zhou weights). *For psh germs u_1, u_2 at o , the following are equivalent:*

- (i) $\nu_\Phi(u_1) = \nu_\Phi(u_2)$ for every local Zhou weight Φ ;
- (ii) $\widehat{u}_1 = \widehat{u}_2$ on the normalized valuative space.

In particular, equality of generalized Lelong numbers against all tame weights implies equality of Lelong numbers on all modifications, giving (4) $\Rightarrow$ (1) in Theorem A of [4].

Bao–Guan–Zhou established weak density of the cone of Zhou valuations and its analytic correspondence [3, Theorems 9.2 and 11.2]. We retain a bound $A(w) \leq A(v)$ in their approximation of a finite-discrepancy valuation v . The valuative Demailly estimate then upgrades finite-ideal approximation to simultaneous approximation on finitely many psh germs. This avoids assuming continuity of arbitrary valuative transforms in the weak topology.

Once flatness is known, the multiplier-ideal separation theorem of [1] also yields [theorem 1.2](#), together with [4, Theorem A]. We give the controlled-density proof here. The argument uses analytic mass comparison, algebraic multiplicities, and Zhou density, with no quasi-monomiality assumption on Zhou valuations.

2. ZHOU WEIGHTS AND ANALYTIC INPUTS

2.1. Conventions and Zhou weights. We exclude the identically $-\infty$ germ from PSH_o . Use $d^c = (\partial - \bar{\partial})/(2\pi i)$, so that $dd^c \log |f| = [\text{div}(f)]$. For a nonzero ideal $I = (f_1, \dots, f_r)$, set

$$\log |I| = \frac{1}{2} \log(|f_1|^2 + \dots + |f_r|^2).$$

Changing generators changes this germ by a bounded function. All comparisons of singularities in this paper are modulo bounded functions. Weights have isolated poles and continuous exponentials. A weight is maximal if

$(dd^c\Phi)^n$ vanishes off o . The masses are the classical generalized Lelong products, not non-pluripolar products. The multiplier ideal is

$$\mathcal{J}(u)_o = \{f \in R : |f|^2 e^{-2u} \text{ is locally integrable at } o\}.$$

We omit the stalk subscript below.

Definition 2.1 ([1]). Let F_0 be a nonzero tuple of holomorphic germs and let ψ_0 be psh, with $|F_0|^2 e^{-2\psi_0}$ locally integrable. A local Zhou weight for this datum is a psh germ Φ satisfying:

- (i) $|F_0|^2 e^{-2\psi_0} |z|^{2N} e^{-2\Phi}$ is integrable for some integer $N > 0$;
- (ii) $|F_0|^2 e^{-2\psi_0} e^{-2\Phi}$ is not integrable;
- (iii) if $\Psi \geq \Phi + O(1)$ is psh and $|F_0|^2 e^{-2\psi_0} e^{-2\Psi}$ is not integrable, then $\Psi = \Phi + O(1)$.

The datum fixes the scale of Φ , not the normalization $w_\Phi(\mathfrak{m}) = 1$. Tame-ness means that a constant $C > 0$ satisfies

$$(2.1) \quad f \in \mathcal{J}(t\Phi) \implies \log |f| \leq (t - C)\Phi + O(1), \quad t > 0.$$

We use the following results of [1, Section 1.1, Proposition 1.7, and Section 1.2]: every local Zhou weight has such a tame maximal representative; there are constants $0 < c < L$ with

$$(2.2) \quad L \log |z| + O(1) \leq \Phi \leq c \log |z| + O(1);$$

and w_Φ is a real valuation, finite on every nonzero germ, positive on $\mathfrak{m} \setminus \{0\}$, and zero on units. For weights related to holomorphic ideals, an algebraic treatment of the correspondence and tameness is given in [3, Theorem 11.2 and Corollary 11.8]. Comparison with $\log |z|$ in (2.2) gives $0 < c^n \leq \tau_\Phi \leq L^n < \infty$.

2.2. Comparison, approximation, and multiplicity. We use Demailly's comparison and approximation theorems [6, 7]; the relative-type attainment statement is discussed in [12] and [4, Section 5.4].

Proposition 2.2. *The following statements hold.*

- (i) *For maximal Φ , one has $u \leq \sigma(u, \Phi)\Phi + O(1)$. The relative type is finite for $u \in \text{PSH}_o$ when (2.2) holds.*
- (ii) *For $t > 0$, Demailly approximation gives $u \leq t^{-1} \log |\mathcal{J}(tu)| + O(1)$.*
- (iii) *If weights $\alpha \leq \beta + O(1)$ have isolated poles, then $\nu_\alpha(u) \geq \nu_\beta(u)$. Residual Monge–Ampère masses obey the same monotonicity. Scaling a weight by $b > 0$ multiplies these two quantities by b^{n-1} and b^n , respectively.*
- (iv) *If I is $\mathfrak{m}$ -primary, then*

$$(dd^c \log |I|)^n \{o\} = e(I; R), \quad \nu_{\log |I|}(\log |f|) = e(I; R/(f)) \quad (0 \neq f \in \mathfrak{m}).$$

For irreducible f , Poincaré–Lelong identifies the second mass in (iv) with $(dd^c \log |I|)^{n-1} \wedge [f = 0] \{o\}$. The analytic multiplicity formula on the possibly singular reduced hypersurface is also a case of [8, Theorem 1.1], with

the same ideal repeated $n - 1$ times. For general f , both sides are additive over its irreducible factors, with their multiplicities.

2.3. The BFJ valuative formulas. Let $\mathcal{V}$ be the compact space of semi-valuations on the completed local ring, centered at o and normalized by $v(\mathfrak{m}) = 1$. Evaluations of nonzero elements may equal $+\infty$ at a semivaluation. Normalized valuations on R extend uniquely to this space by $\mathfrak{m}$ -adic truncation. Write $\mathcal{V}_{\text{qm}}$ for the quasi-monomial points. For an ideal $I = (f_i)$, $v(I) = \min_i v(f_i)$; in particular $v(IJ) = v(I) + v(J)$. At a divisorial valuation ord_E , $-\widehat{u}(\text{ord}_E)$, interpreted homogeneously, is the generic Lelong number of the pullback along E .

The valuative transform $\widehat{u}$ is a nonpositive formal psh function. It satisfies

$$\widehat{\log |I|}(v) = -v(I), \quad \widehat{bu} = b\widehat{u} \quad (b > 0), \quad \widehat{\max(u_1, u_2)} = \max(\widehat{u}_1, \widehat{u}_2).$$

For an unnormalized real valuation w , put

$$(2.3) \quad w[u] = w(\mathfrak{m})(-\widehat{u}(w/w(\mathfrak{m}))).$$

In particular, $w[\log |I|] = w(I)$.

Theorem 2.3 ([4]). *Let Φ be a tame maximal weight and $g = \widehat{\Phi}$. Then g is bounded, continuous, and strictly negative on $\mathcal{V}$, and*

$$(2.4) \quad \nu_{\Phi}(u) = \int_{\mathcal{V}} -\widehat{u} d\text{MA}(g),$$

$$(2.5) \quad \sigma(u, \Phi) = \inf_{v \in \mathcal{V}} \frac{-\widehat{u}(v)}{-g(v)}.$$

Every formal psh function h admits a sequence $h_j = c_j \log |I_j|$ decreasing to h at every point of $\mathcal{V}$, where $c_j > 0$ and I_j are primary ideals. Finally, its values on all of $\mathcal{V}$ are determined by its values on $\mathcal{V}_{\text{qm}}$:

$$(2.6) \quad h(v) = \inf_{\pi} h(r_{\pi}(v)).$$

Here $r_{\pi}(v)$ is the quasi-monomial retraction associated to a sufficiently high simple modification π .

The sources are Theorem B, Theorem 5.12, Proposition 3.13, and Definition 2.8 of [4], respectively. Boundedness and strict negativity follow from (2.2). Continuity follows from the tame multiplier-ideal approximation [4, Lemma 5.9]: the continuous transforms of primary ideal weights converge uniformly, since their errors are bounded by a constant times $(-g)/m$. Primary ideals in the formal approximation may be regarded as analytic ideals: if $\mathfrak{m}^N \subset I$, truncating generators to polynomials modulo $\mathfrak{m}^N$ does not change the completed ideal or its valuation function.

3. VALUATION IDEALS ON HYPERSURFACES

Let w be any centered real valuation of $R = \mathbb{C}\{z_1, \dots, z_n\}$. Set $\lambda = w(\mathfrak{m}) > 0$ and

$$\mathfrak{a}_t = \{f : w(f) \geq t\} \quad (t > 0), \quad \mathfrak{a}_t = R \quad (t \leq 0), \quad H(t) = \ell_R(R/\mathfrak{a}_t).$$

The family is decreasing and graded: $\mathfrak{a}_s \mathfrak{a}_t \subset \mathfrak{a}_{s+t}$. Moreover $\mathfrak{m}^{\lceil t/\lambda \rceil} \subset \mathfrak{a}_t \subset \mathfrak{m}$ for $t > 0$, so its positive-index ideals are primary.

Theorem 3.1 (Cutkosky). *Let $(S, \mathfrak{n})$ be an analytically unramified equicharacteristic local ring of dimension $d > 0$, with perfect residue field. For a graded family $I_\bullet$ of $\mathfrak{n}$ -primary ideals, both limits below exist and*

$$\lim_{m \rightarrow \infty} \frac{d! \ell_S(S/I_m)}{m^d} = \lim_{m \rightarrow \infty} \frac{e(I_m; S)}{m^d}.$$

This is [5, Theorems 5.8 and 6.5]. Reduced complex analytic local rings are excellent with reduced completion, so the theorem applies to R and to $R/(f)$ for irreducible f . In particular,

$$(3.1) \quad \text{vol}(w) = \lim_{m \rightarrow \infty} \frac{n! H(m)}{m^n} = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m; R)}{m^n}.$$

The limit is finite; positivity is not assumed here. Monotonicity of H extends its asymptotic formula from integer to real indices by comparison with the floor and ceiling.

Proposition 3.2 (Hypersurface-volume formula). *For $0 \neq f \in \mathfrak{m}$,*

$$\lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} = w(f) \text{vol}(w).$$

No finite-generation or quasi-monomiality hypothesis is required.

Proof. First suppose f is irreducible. Put $s = w(f) > 0$ and $S = R/(f)$. For every real t and $h \in R$,

$$h \in (\mathfrak{a}_t : f) \iff w(fh) = s + w(h) \geq t \iff h \in \mathfrak{a}_{t-s}.$$

Thus multiplication by f gives the exact sequence

$$(3.2) \quad 0 \longrightarrow R/\mathfrak{a}_{t-s} \xrightarrow{\cdot f} R/\mathfrak{a}_t \longrightarrow R/(\mathfrak{a}_t + (f)) \longrightarrow 0.$$

Consequently

$$(3.3) \quad L_f(t) := \ell_S(S/\mathfrak{a}_t S) = H(t) - H(t-s).$$

The ideals $\mathfrak{a}_m S$ are a graded primary family in the reduced analytic local ring S of dimension $n-1$. By [theorem 3.1](#), there is a finite number B_f such that

$$(3.4) \quad \lim_{m \rightarrow \infty} \frac{(n-1)! L_f(m)}{m^{n-1}} = B_f = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m; S)}{m^{n-1}}.$$

Since L_f is increasing, its asymptotic formula again holds for real indices. Evaluating (3.3) at $t = s, 2s, \dots, qs$ and using $H(0) = 0$, we obtain

$$\begin{aligned} H(qs) &= \sum_{j=1}^q L_f(js) \\ &= \frac{B_f s^{n-1}}{(n-1)!} \sum_{j=1}^q j^{n-1} + o(q^n) \\ &= \frac{B_f s^{n-1}}{n!} q^n + o(q^n). \end{aligned}$$

Here $\sum_{j \leq q} o(j^{n-1}) = o(q^n)$. On the other hand, (3.1) yields

$$H(qs) = \frac{\text{vol}(w)s^n}{n!} q^n + o(q^n).$$

Comparison gives $B_f = s \text{vol}(w)$, proving the irreducible case.

For general f , unique factorization in R gives $f = a \prod_{i=1}^r f_i^{b_i}$, with a a unit and f_i pairwise nonassociate irreducibles. The associativity formula for multiplicity [10, Chapter 11] gives

$$e(\mathfrak{a}_m; R/(f)) = \sum_{i=1}^r b_i e(\mathfrak{a}_m; R/(f_i)).$$

Indeed, $R_{(f_i)}$ is a discrete valuation ring and $\ell((R/(f))_{(f_i)}) = b_i$. Divide by m^{n-1} and use the irreducible case together with $w(f) = \sum_i b_i w(f_i)$. $\square$

Remark 3.3. The ambient estimate $H(t) = \text{vol}(w)t^n/n! + o(t^n)$ does not by itself determine a fixed-width difference of order t^{n-1} . The existence of the hypersurface limit is essential. Factorization treats the nonreduced case without applying theorem 3.1 to a nonreduced ring.

4. FLATNESS OF ZHOU WEIGHTS

Fix a tame maximal representative Φ of a local Zhou weight and put $w = w_\Phi$.

4.1. The valuation-ideal approximation.

Lemma 4.1. *There is an integer $C > 0$ independent of m such that, with $\Psi_m = m^{-1} \log |\mathfrak{a}_m(w)|$,*

$$(4.1) \quad (1 + C/m)\Phi \leq \Psi_m \leq \Phi \quad \text{modulo bounded functions.}$$

Proof. Choose an integer C at least the tameness constant. Equation (2.1) implies $\mathcal{J}((m+C)\Phi) \subset \mathfrak{a}_m(w)$. Demailly approximation and inclusion of ideals therefore give

$$(m+C)\Phi \leq \log |\mathcal{J}((m+C)\Phi)| + O(1) \leq \log |\mathfrak{a}_m(w)| + O(1).$$

For the other inequality, each generator f of $\mathfrak{a}_m(w)$ satisfies $\sigma(\log |f|, \Phi) \geq m$. Attainment of relative type implies $\log |f| \leq m\Phi + O(1)$. Taking the maximum over finitely many generators and dividing by m proves (4.1). $\square$

Proposition 4.2. *One has $\tau_\Phi = \text{vol}(w) > 0$ and*

$$(4.2) \quad \nu_\Phi(\log |f|) = \text{vol}(w)w(f) \quad \text{for every } f \in R \setminus \{0\}.$$

Proof. By proposition 2.2 and lemma 4.1,

$$(4.3) \quad \tau_\Phi \leq \frac{e(\mathfrak{a}_m(w); R)}{m^n} \leq (1 + C/m)^n \tau_\Phi.$$

The middle quantity is the residual mass of Ψ_m . Its limit is $\text{vol}(w)$ by (3.1), so the squeeze proves $\tau_\Phi = \text{vol}(w) > 0$, by (2.2).

For irreducible f , the hypersurface multiplicity formula gives

$$\nu_{\Psi_m}(\log |f|) = \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}}.$$

Comparison in the weight variable has degree $n - 1$, hence

$$(4.4) \quad \nu_\Phi(\log |f|) \leq \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} \leq (1 + C/m)^{n-1} \nu_\Phi(\log |f|).$$

By proposition 3.2, the middle term tends to $w(f) \text{vol}(w)$. This proves (4.2) for irreducible f . Factorization and additivity of both $dd^c \log |f|$ and $w(f)$ give the general case. A unit contributes zero to both sides. $\square$

4.2. From holomorphic tests to all psh germs.

Lemma 4.3 (Holomorphic testing). *Let Φ be tame and maximal, and suppose*

$$\nu_\Phi(\log |f|) = \tau_\Phi \sigma(\log |f|, \Phi) \quad \text{for every nonzero } f \in R.$$

Then there is a unique $v_0 \in \mathcal{V}$ such that

$$(4.5) \quad \text{MA}(g) = \frac{\tau_\Phi}{-g(v_0)} \delta_{v_0}, \quad \frac{\nu_\Phi(u)}{\tau_\Phi} = \sigma(u, \Phi) = \frac{-\widehat{u}(v_0)}{-g(v_0)}, \quad g = \widehat{\Phi},$$

for every psh germ u .

Proof. Applying (2.4) to $u = \Phi$ gives $\int_{\mathcal{V}} (-g) d\text{MA}(g) = \tau_\Phi$. Thus

$$dP = \frac{-g}{\tau_\Phi} d\text{MA}(g)$$

is a Radon probability measure. For a nonzero holomorphic germ f , set $R_f(v) = v(f)/(-g(v))$ and $a_f = \sigma(\log |f|, \Phi)$. The hypotheses and theorem 2.3 give

$$\inf_{\mathcal{V}} R_f = a_f = \int_{\mathcal{V}} R_f dP.$$

The nonnegative function $R_f - a_f$ therefore has integral zero. It is continuous with values in $[0, +\infty]$, because evaluation of f is continuous in the valuation topology and $-g$ is continuous and bounded away from zero. If $R_f(v) > a_f$

at a point of $\text{supp } P$, some neighborhood of that point has positive P -mass and $R_f - a_f \geq \varepsilon > 0$ there, a contradiction. Consequently

$$(4.6) \quad \frac{v(f)}{-g(v)} = a_f \quad \text{for every } v \in \text{supp } P \text{ and every } f \neq 0.$$

The support is nonempty. If v_1, v_2 belong to it, then $v_1/(-g(v_1))$ and $v_2/(-g(v_2))$ agree on every holomorphic germ. Taking their minima on the coordinate generators of $\mathfrak{m}$ and using $v_i(\mathfrak{m}) = 1$ shows $-g(v_1) = -g(v_2)$, and therefore equality on R . Polynomial truncations determine each normalized semivaluation on the completion: for a formal series F and its degree- N truncation F_N , $v(F - F_N) \geq N + 1$. Hence $v_1 = v_2$ on the completion as well. By Radon regularity, $P = \delta_{v_0}$ for one point v_0 .

It remains to identify the relative type for an arbitrary u . By (4.6) and the definition of the infimum,

$$\frac{v_0(f)}{-g(v_0)} \leq \frac{v(f)}{-g(v)} \quad \text{for every } v \in \mathcal{V} \text{ and every } f \neq 0.$$

Taking the minimum over generators proves this inequality for any primary ideal, and positive scaling preserves it. Choose $h_j = c_j \log |I_j| \downarrow \hat{u}$ as in [theorem 2.3](#). For each fixed v , pass to the limit in

$$\frac{-h_j(v_0)}{-g(v_0)} \leq \frac{-h_j(v)}{-g(v)}.$$

It follows that v_0 attains the infimum in (2.5) for u . Integration against $P = \delta_{v_0}$ gives the generalized Lelong number in (4.5). The measure formula follows from the definition of P . $\square$

Proof of [theorem 1.1](#). By [proposition 4.2](#), the hypothesis of [lemma 4.3](#) holds. Hence $\nu_\Phi(u) = \tau_\Phi \sigma(u, \Phi)$ for all u . On holomorphic germs, that lemma also gives

$$w(f) = \frac{v_0(f)}{-g(v_0)}.$$

Taking minima on generators of $\mathfrak{m}$ yields $\lambda = 1/(-g(v_0))$, and therefore $v_0 = w/\lambda$. Since $\tau_\Phi = \text{vol}(w)$, all formulas in (1.1) and (1.2) follow. $\square$

Since $\mathfrak{a}_t(bv) = \mathfrak{a}_{t/b}(v)$, one has $\text{vol}(bv) = b^{-n} \text{vol}(v)$. Thus the normalized formulas are

$$\tau_\Phi = \lambda^{-n} \text{vol}(v), \quad \text{MA}(\hat{\Phi}) = \lambda^{1-n} \text{vol}(v) \delta_v.$$

The squeezes (4.3) and (4.4) give relative errors $O(m^{-1})$ in the corresponding multiplicity limits.

5. CONTROLLED DENSITY OF ALGEBRAIC ZHOU VALUATIONS

5.1. Algebraic Zhou valuations and analytic correspondence. We apply [3] to $X = \text{Spec } R$, which is regular, excellent and Noetherian of characteristic zero. All valuations and ideals remain on the analytic local ring R .

For a graded sequence $I_\bullet$, write

$$w(I_\bullet) = \lim_m \frac{w(I_m)}{m} = \inf_m \frac{w(I_m)}{m}, \quad \text{lct}^q(I_\bullet) = \sup_m m c_o^q(\log |I_m|),$$

where $c_o^q(u) = \sup\{t \geq 0 : |q|^2 e^{-2tu} \text{ is locally integrable}\}$. For ideals on R , this analytic threshold agrees with the algebraic jumping number; see [3, Proposition 11.1]. The valuative formula gives, in particular,

$$(5.1) \quad \text{lct}^q(I_\bullet) \leq \frac{A(v) + v(q)}{v(I_\bullet)} \quad \text{when } v(I_\bullet) > 0, A(v) < \infty.$$

Here A is log discrepancy [11, Section 5]: $A(\text{ord}_E) = 1 + \text{ord}_E(K_{Y/X})$, it extends linearly on each quasi-monomial cone, and on arbitrary valuations it is the supremum over quasi-monomial retractions. In particular,

$$A(bv) = bA(v), \quad A(r_\pi(v)) \leq A(v).$$

A centered nontrivial valuation of finite discrepancy has $A(v) > 0$.

Definition 5.1 ([3, Definition 3.2]). An algebraic Zhou valuation related to a nonzero ideal q is a maximal element, for pointwise order on R , among valuations w satisfying $\text{lct}^q(\mathbf{a}_\bullet(w)) \leq 1$. We write $\mathcal{Z}$ for the cone of positive multiples of algebraic Zhou valuations centered at o .

Proposition 5.2 (Bao–Guan–Zhou). (i) *If $\tilde{w}$ is an algebraic Zhou valuation related to q_0 , then*

$$(5.2) \quad A(\tilde{w}) = 1 - \tilde{w}(q_0) < \infty.$$

- (ii) *If v is centered and $C = \text{lct}^q(\mathbf{a}_\bullet(v))$ is finite and positive, there is an algebraic Zhou valuation $\tilde{w}$ related to q such that $C\tilde{w} \geq v$.*
- (iii) *A valuation on R centered at o is an algebraic Zhou valuation related to q_0 if and only if it equals w_Φ for an analytic Zhou weight Φ related to $|q_0|^2$.*

Part (i) is [3, Theorem 4.1]; part (ii) follows by applying [3, Corollary 3.4] to v/C , as in the proof of its Lemma 9.1; part (iii) is its Theorem 11.2. For (ii), the scaling identity $\mathbf{a}_t(v/C) = \mathbf{a}_{Ct}(v)$ makes the threshold of v/C equal to one. Also $\tilde{w} \geq v/C$ implies $\tilde{w}(\mathfrak{m}) > 0$, so the analytic correspondence applies.

Consequently every $w \in \mathcal{Z}$ has finite log discrepancy and can be written $w = bw_\Phi$ with $b > 0$ and Φ a local Zhou weight. By [theorem 1.1](#),

$$(5.3) \quad w[u] = b\sigma(u, \Phi).$$

5.2. A bounded-discrepancy refinement of density.

Proposition 5.3. *Let v be a centered valuation with $A = A(v) < \infty$. Given nonzero ideals $I_1, \dots, I_r$ and an integer $k \geq 1$, there exists $w \in \mathcal{Z}$ such that*

$$(5.4) \quad w \geq v, \quad 0 < A(w) \leq A, \quad 0 \leq w(I_i) - v(I_i) \leq A/k \quad (1 \leq i \leq r).$$

The same choice can be required to satisfy $0 \leq w(\mathfrak{m}) - v(\mathfrak{m}) \leq A/k$.

Proof. Put $\mathbf{q} = \mathbf{m}I_1 \cdots I_r$ and $C_k = \text{lct}^{\mathbf{q}^k}(\mathbf{a}_\bullet(v))$. First, $v(\mathbf{a}_\bullet(v)) = 1$: $v(\mathbf{a}_m(v)) \geq m$, while, for any fixed germ f of value $s = v(f) > 0$, the power $f^{\lceil m/s \rceil}$ belongs to $\mathbf{a}_m(v)$ and has value less than $m + s$. Thus (5.1) gives

$$(5.5) \quad C_k \leq A + kv(\mathbf{q}) < \infty.$$

Moreover, $\mathbf{m}^{\lceil m/v(\mathbf{m}) \rceil} \subset \mathbf{a}_m(v)$, so the ordinary threshold of $\mathbf{a}_m(v)$ is at least $n/\lceil m/v(\mathbf{m}) \rceil$. A holomorphic multiplier can only increase the threshold. Taking the supremum of m times these lower bounds gives $C_k \geq nv(\mathbf{m}) > 0$.

By proposition 5.2(ii), there is $\tilde{w}_k$ related to $\mathbf{q}^k$ with $w_k := C_k \tilde{w}_k \geq v$. Homogeneity and (5.2) give the exact relation

$$(5.6) \quad A(w_k) = C_k - kw_k(\mathbf{q}).$$

Combining (5.5), $w_k \geq v$, and $A(w_k) > 0$ gives

$$0 < A(w_k) \leq A - k(w_k(\mathbf{q}) - v(\mathbf{q})).$$

Consequently

$$0 < A(w_k) \leq A, \quad 0 \leq w_k(\mathbf{q}) - v(\mathbf{q}) \leq A/k.$$

Since

$$w_k(\mathbf{q}) - v(\mathbf{q}) = w_k(\mathbf{m}) - v(\mathbf{m}) + \sum_{i=1}^r (w_k(I_i) - v(I_i)),$$

and every summand is nonnegative, each is at most A/k . Take $w = w_k$. $\square$

Remark 5.4. The construction is that of [3, Lemma 9.1 and Theorem 9.2], with the additional bound supplied by (5.6) retained. The weak density theorem itself is already established there; an earlier analytic version appears in [2, Theorems 1.13–1.14]. The bound on A makes the following psh approximation uniform.

5.3. Uniform multiplier-ideal approximation.

Lemma 5.5. *Let $u \in \text{PSH}_o$ and let v be centered with $A(v) < \infty$. For every integer $m \geq 1$,*

$$(5.7) \quad 0 \leq v[u] - \frac{v(\mathcal{J}(mu))}{m} \leq \frac{A(v)}{m}.$$

In particular, this holds for every $v \in \mathcal{Z}$.

Proof. For quasi-monomial v , this is the valuative Demailly estimate [4, Equation (5.3), Section 5.2], namely

$$v(\mathcal{J}(mu)) \leq mv[u] \leq v(\mathcal{J}(mu)) + A(v).$$

The left inequality follows from $u \leq m^{-1} \log |\mathcal{J}(mu)| + O(1)$. For $v = \text{ord}_E$, write $a = v(f)$, $\gamma = v[u]$, and $k_E = \text{ord}_E(K_{Y/X})$. At a general point of $E = (t = 0)$, integrability of the pullback of $|f|^2 e^{-2mu}$ forces $a + k_E - m\gamma > -1$, since the remaining psh potential is bounded above. Thus $mv[u] \leq v(f) + A(v)$ for $f \in \mathcal{J}(mu)$. Taking the minimum over generators proves the

estimate; homogeneity and continuity on quasi-monomial simplices extend it by rational approximation [4, Proposition 5.6].

For general v , Demailly approximation still gives the left inequality. The homogeneous quasi-monomial retractions satisfy

$$r_\pi(v) \leq v \text{ on } R, \quad A(r_\pi(v)) \leq A(v), \quad v[u] = \sup_\pi r_\pi(v)[u].$$

The first two properties are part of the retraction and discrepancy formalism of [11, Sections 4–5]; the last is (2.6) with the sign reversed. Applying the quasi-monomial estimate gives

$$mr_\pi(v)[u] \leq r_\pi(v)(\mathcal{J}(mu)) + A(r_\pi(v)) \leq v(\mathcal{J}(mu)) + A(v).$$

Taking the supremum proves (5.7). Its right side is finite because $\mathcal{J}(mu)$ is a nonzero ideal and v is a real valuation. Thus $v[u]$ is finite as well. $\square$

Lemma 5.6 (Order and evaluation). *If w, v are centered, of finite log discrepancy, and $w \geq v$ on R , then $w[u] \geq v[u]$ for every $u \in \text{PSH}_o$.*

Proof. For every integer m , one has $w(\mathcal{J}(mu)) \geq v(\mathcal{J}(mu))$. By lemma 5.5, dividing by m and taking the limit gives $w[u] \geq v[u]$. $\square$

Theorem 5.7 (Finite-psh approximation by Zhou valuations). *Let $v(\mathfrak{m}) = 1$ and $A = A(v) < \infty$, and fix finitely many psh germs $u_1, \dots, u_r$. There are $w_j \in \mathcal{Z}$, $j \geq 1$, with*

$$(5.8) \quad w_j \geq v, \quad A(w_j) \leq A, \quad 1 \leq w_j(\mathfrak{m}) \leq 1 + A/j,$$

$$(5.9) \quad 0 \leq w_j[u_i] - v[u_i] \leq \frac{A}{j} + \frac{A}{j^2} \quad (1 \leq i \leq r).$$

In particular $v_j = w_j/w_j(\mathfrak{m})$ is a normalized Zhou ray and $v_j[u_i] \rightarrow v[u_i]$ simultaneously for all i .

Proof. For each j , apply proposition 5.3 with $k = j$ and $I_i = \mathcal{J}(ju_i)$. These ideals are nonzero; they may equal R , which causes no change in the product construction. We obtain (5.8) and

$$0 \leq w_j(\mathcal{J}(ju_i)) - v(\mathcal{J}(ju_i)) \leq A/j.$$

Applying lemma 5.5 to both valuations gives

$$\begin{aligned} w_j[u_i] &\leq \frac{w_j(\mathcal{J}(ju_i))}{j} + \frac{A(w_j)}{j} \\ &\leq \frac{v(\mathcal{J}(ju_i))}{j} + \frac{A}{j^2} + \frac{A}{j} \\ &\leq v[u_i] + \frac{A}{j} + \frac{A}{j^2}. \end{aligned}$$

The lower bound is lemma 5.6. For normalization, set $d_j = w_j(\mathfrak{m}) - 1$ and $e_{i,j} = w_j[u_i] - v[u_i]$. Since $v[u_i] < \infty$,

$$|v_j[u_i] - v[u_i]| = \left| \frac{e_{i,j} - d_j v[u_i]}{1 + d_j} \right| \leq \frac{A}{j} \left(1 + \frac{1}{j} + v[u_i] \right) \rightarrow 0. \quad \square$$

6. VALUATIVE SEPARATION AND THE BFJ CONVERSE

Proof of theorem 1.2. Assume first that $\nu_\Phi(u_1) = \nu_\Phi(u_2)$ for every local Zhou weight. By theorem 1.1, division by the fixed positive mass τ_Φ gives $\sigma(u_1, \Phi) = \sigma(u_2, \Phi)$. Equation (5.3) therefore implies

$$(6.1) \quad w[u_1] = w[u_2] \quad \text{for every } w \in \mathcal{Z}.$$

Fix $v \in \mathcal{V}_{\text{qm}}$. Apply theorem 5.7 to the pair (u_1, u_2) to obtain the same sequence w_j for both germs. Using (6.1) and (5.9), we find

$$v[u_1] = \lim_j w_j[u_1] = \lim_j w_j[u_2] = v[u_2].$$

Thus $\hat{u}_1 = \hat{u}_2$ on $\mathcal{V}_{\text{qm}}$. For arbitrary $\mu \in \mathcal{V}$, the retraction formula (2.6) gives

$$\hat{u}_1(\mu) = \inf_{\pi} \hat{u}_1(r_{\pi}(\mu)) = \inf_{\pi} \hat{u}_2(r_{\pi}(\mu)) = \hat{u}_2(\mu).$$

This proves (ii). Conversely, if the valuative transforms agree, their integrals against $\text{MA}(\hat{\Phi})$ agree by (2.4). This gives (i).

Finally, every Zhou weight has a tame maximal representative, which is an allowed test in condition (4) of Theorem A of [4]. The implication just proved yields equality of the valuative transforms. This is equivalent to equality of Lelong numbers of the pullbacks at every point of every modification above o , namely condition (1); see [4, Sections 5.1 and 5.6]. $\square$

Corollary 6.1 (Order detection). *For $u_1, u_2 \in \text{PSH}_o$, the following are equivalent:*

$$\nu_\Phi(u_1) \geq \nu_\Phi(u_2) \quad \text{for every local Zhou weight } \Phi; \quad \hat{u}_1 \leq \hat{u}_2 \quad \text{on } \mathcal{V}.$$

Proof. Divide each test inequality by $\tau_\Phi > 0$ and use theorem 1.1. This gives $w[u_1] \geq w[u_2]$ on $\mathcal{Z}$. The simultaneous approximation in theorem 5.7 preserves this inequality at each $v \in \mathcal{V}_{\text{qm}}$; applying (2.6) extends the inequality of transforms to $\mathcal{V}$. The converse follows by integrating $-\hat{u}_1 \geq -\hat{u}_2$ against the positive measure $\text{MA}(\hat{\Phi})$. $\square$

Corollary 6.2 (Multiplier ideals). *Under either condition of theorem 1.2, $\mathcal{J}(tu_1) = \mathcal{J}(tu_2)$ for every $t > 0$.*

Proof. Theorem A of [4] gives equality of $\mathcal{J}_+(tu_i)$. The strong openness theorem of Guan–Zhou [9] identifies $\mathcal{J}_+(tu_i)$ with $\mathcal{J}(tu_i)$. $\square$

In dimension one, the same conclusions follow directly from $\Phi = a \log |z| + O(1)$: its residual mass is a , and both normalized generalized Lelong number and relative type equal $\nu(u, o)/a$.

REFERENCES

1. Shijie Bao, Qi'an Guan, Zhitong Mi, and Zheng Yuan, *Tame maximal weights, relative types and valuations*, Adv. Math. **477** (2025), 110364.
2. Shijie Bao, Qi'an Guan, and Zheng Yuan, *Zhou valuations and jumping numbers*, arXiv:2311.06565v2, 2024.

3. Shijie Bao, Qi'an Guan, and Lin Zhou, *Algebraic Zhou valuations*, arXiv:2505.19451v3, 2025.
4. Sébastien Boucksom, Charles Favre, and Mattias Jonsson, *Valuations and plurisubharmonic singularities*, Publ. Res. Inst. Math. Sci. **44** (2008), no. 2, 449–494.
5. Steven Dale Cutkosky, *Multiplicities associated to graded families of ideals*, Algebra Number Theory **7** (2013), no. 9, 2059–2083.
6. Jean-Pierre Demailly, *Nombres de Lelong généralisés, théorèmes d'intégralité et d'analyticité*, Acta Math. **159** (1987), no. 3–4, 153–169.
7. ———, *Monge–Ampère operators, Lelong numbers and intersection theory*, Complex Analysis and Geometry (Vincenzo Ancona and Alessandro Silva, eds.), Plenum Press, New York, 1993, pp. 115–193.
8. Fusheng Deng, Yinji Li, Qunhuan Liu, and Xiangyu Zhou, *Multiplicities and modifications, and singularities associated to blowing down negative vector bundles*, arXiv:2409.09407v1, 2024.
9. Qi'an Guan and Xiangyu Zhou, *A proof of Demailly's strong openness conjecture*, Ann. of Math. (2) **182** (2015), no. 2, 605–616.
10. Craig Huneke and Irena Swanson, *Integral closure of ideals, rings, and modules*, London Mathematical Society Lecture Note Series, vol. 336, Cambridge University Press, 2006.
11. Mattias Jonsson and Mircea Mustață, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.
12. Alexander Rashkovskii, *Relative types and extremal problems for plurisubharmonic functions*, Int. Math. Res. Not. (2006), Art. ID 76283, 26 pp.
13. ———, *An extremal problem for generalized Lelong numbers*, Math. Z. **266** (2010), no. 2, 345–362.

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author proposed the research questions and the organization around Zhou weights, their generalized Lelong numbers, and the density of Zhou valuations. The hypersurface-volume proof, its application to Zhou weights, and the controlled-density argument presented here were developed and written with GPT-6 Astra. Background formulations in the preceding draft were prepared with GPT-5.6-sol (high). The results of other authors used in these arguments are identified by citations. This manuscript remains subject to the author's mathematical verification before submission.

GRADUATE SCHOOL OF SCIENCE, OSAKA METROPOLITAN UNIVERSITY, OSAKA, JAPAN
 Email address: shijie.bao@omu.ac.jp